

Classical Simulation of Grover's Algorithm and Its Limitations Compared to Silicon-Based Quantum Hardware

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Abstract

This project investigates the implementation of Grover's quantum search algorithm using classical simulation tools and compares the results to a recent experimental realization on a silicon-based quantum processor (Thorvaldson et al., 2025). Grover's algorithm provides a quadratic speedup for searching an unsorted database of N items, requiring only $\sqrt{N}$ steps compared to the N steps needed classically, making it one of the most practically significant quantum algorithms. Using Python, Qiskit, and the AerSimulator backend, Grover's algorithm was simulated across systems ranging from 2 to 22 qubits. Metrics collected included runtime, success probability, circuit depth, and total gate count. The simulation achieved near-perfect success probabilities across all qubit counts, frequently reaching 100%, which reflects the noise-free, ideal conditions of classical simulation. However, runtime grew sharply with qubit count, from under one second at 2 qubits to over 1,191 seconds at 22 qubits, demonstrating the exponential cost of classically representing quantum states. These results are compared to the four-qubit silicon-based processor of Thorvaldson et al. (2025), which achieved approximately 95% success probability under realistic hardware noise. The findings confirm that while classical simulation is a powerful tool for learning and verification, it cannot scale to the regime where quantum hardware offers a genuine computational advantage.

1. Introduction

Quantum computing exploits the principles of quantum mechanics to solve certain computational problems far more efficiently than any classical computer. Unlike a classical bit, which is restricted to a value of exactly 0 or 1, a quantum bit (qubit) can exist in a superposition of both states simultaneously. This property, combined with quantum interference and entanglement, allows quantum algorithms to explore many possible solutions in parallel and amplify the probability of correct answers through carefully designed interference patterns.

One of the most celebrated demonstrations of quantum advantage is Grover's search algorithm (Grover, 1996; Nielsen & Chuang, 2010). Given an unstructured database of N items with a single marked solution, a classical computer requires on average $N/2$ queries to find it. Grover's algorithm, by contrast, finds the solution in approximately $\sqrt{N}$ steps, a quadratic speedup that is provably optimal for quantum search. For example, searching a database of one million items classically requires up to 500,000 queries; Grover's algorithm needs roughly 785. While a quadratic speedup is modest compared to the exponential advantage of algorithms like Shor's, Grover's algorithm is broadly applicable because database search is a fundamental primitive in computer science.

The core mechanism of Grover's algorithm involves two repeated operations: an oracle that marks the target state by flipping its phase, and a diffusion operator (also called the Grover diffusion operator or inversion about the mean) that amplifies the amplitude of the marked state. After approximately $(\pi/4)\sqrt{N}$ iterations of this oracle-diffusion cycle, the target state has near-unit probability of being measured. This wave-like amplification is entirely quantum mechanical and has no efficient classical analogue.

Recent experimental work has demonstrated Grover's algorithm on real quantum hardware. Thorvaldson et al. (2025) implemented the algorithm on a four-qubit silicon-based quantum processor, achieving

approximately 95% success probability. Silicon quantum processors are notable because they leverage existing semiconductor manufacturing infrastructure, potentially offering a practical path to scalable quantum computing. Despite the impressive result, their system was limited to four qubits due to challenges including qubit decoherence (the tendency of quantum states to lose their quantum properties over time due to environmental interaction), gate infidelity (imperfect quantum operations), and the difficulty of fabricating and controlling larger arrays of silicon qubits.

The goal of this project is to simulate Grover's algorithm classically using Qiskit and the AerSimulator, scaling from 2 to 22 qubits, and to analyze how performance metrics evolve with system size. By comparing these simulation results to the Thorvaldson et al. (2025) hardware experiment, this project illuminates the fundamental difference between ideal simulation and noisy physical quantum hardware, and explains why classical simulation, despite appearing fast at small scales, becomes exponentially prohibitive as qubit count grows.

2. Methods

All simulations were implemented in Python using Qiskit, an open-source quantum computing framework developed by IBM, and executed using the AerSimulator backend, which runs quantum circuits on a classical computer. The code was developed and run in Visual Studio Code.

For each qubit count n from 2 to 22, the following procedure was carried out. First, a random target bitstring of length n was generated using Python's random module, ensuring that no trivially easy targets (such as the all-zeros string) were consistently selected. Second, a Boolean oracle expression was constructed from this target: each bit position contributed a variable ($a, b, c, \dots$) that was either affirmed or negated depending on whether the corresponding target bit was 1 or 0. For example, a target of 101 yielded the expression $a \& \sim b \& c$. This Boolean expression was then compiled into a Qiskit PhaseOracle, which applies a phase flip of -1 to the target state and leaves all other states unchanged, exactly the oracle required by Grover's algorithm.

The number of Grover iterations was set according to the theoretically optimal formula $k \approx (\pi/4)\sqrt{2^n}$, where 2^n is the size of the search space. This value maximizes the probability of measuring the correct state. The full Grover circuit was constructed using Qiskit's Grover and AmplificationProblem classes, transpiled for the AerSimulator, and executed with 1,024 measurement shots. Success probability was defined as the fraction of shots that returned the correct target state.

The following metrics were recorded for each qubit count: total runtime (wall-clock time in seconds), success probability, circuit depth (the number of sequential gate layers), and gate count (total number of quantum gates). For qubit counts of 4 or fewer, the full transpiled circuit diagram was printed to facilitate manual inspection and verification.

To provide a real-hardware comparison, a literature review was conducted focusing on Grover's algorithm implementations on physical quantum processors. The primary reference is Thorvaldson et al. (2025), which describes a four-qubit implementation on a silicon spin-qubit processor. Additional background was drawn from the canonical textbook by Nielsen & Chuang (2010) and the Qiskit documentation. Literature was accessed through standard academic search tools.

3. Results

Table 1 presents the complete simulation results for all qubit counts from 2 to 22. Figure descriptions below summarize the three key trends observed.

Table 1. Grover's Algorithm Simulation Results (2–22 Qubits)

Qubits	States (N)	Iterations	Circuit Depth	Gate Count	Runtime (s)	Success Prob.
2	4	1	6	10	0.95	1.000
3	8	2	10	22	0.28	0.956
4	16	3	14	38	0.27	0.960
6	64	6	26	96	0.27	0.995
8	256	12	50	232	0.31	1.000
10	1,024	25	102	570	0.39	1.000
12	4,096	50	202	1,324	0.58	1.000
14	16,384	100	402	3,028	4.07	1.000
16	65,536	201	806	6,865	15.70	1.000
18	262,144	402	1,610	15,311	45.52	1.000
20	1,048,576	804	3,218	33,808	214.33	1.000
22	4,194,304	1,608	6,434	74,011	1191.08	1.000

Selected qubit counts shown. Full data available from simulation output.

Success probability was consistently high throughout the entire experiment. For most qubit counts, the simulation returned the correct answer with probability 1.000 (100%). The only exceptions were 3 qubits (95.6%), 4 qubits (96.0%), and 6 qubits (99.5%), small deviations attributable to statistical sampling noise from 1,024 measurement shots rather than any algorithmic error. This near-perfect performance is expected in an ideal, noise-free simulation environment.

Runtime exhibited the most dramatic scaling behavior. For systems of up to approximately 12 qubits, runtime remained comfortably below one second. At 14 qubits it reached 4.07 seconds. At 16 qubits: 15.70 seconds. At 18 qubits: 45.52 seconds. At 20 qubits: 214.33 seconds. At 22 qubits: 1,191.08 seconds (approximately 20 minutes). This growth is consistent with exponential scaling, each additional qubit doubles the size of the quantum state vector that the classical computer must maintain in memory and process.

Circuit depth and gate count also scaled super-linearly. Circuit depth grew from 6 at 2 qubits to 6,434 at 22 qubits, and gate count grew from 10 to 74,011 over the same range. This growth is driven by two compounding factors: the number of Grover iterations required grows as $\sqrt{N}$ (itself exponential in n), and the complexity of the oracle circuit increases with the number of qubits being conditioned upon.

Compared to the Thorvaldson et al. (2025) silicon hardware experiment, the 4-qubit simulation achieved a success probability of 96.0%, slightly higher than the reported hardware result of approximately 95%. This small gap is directly attributable to the absence of noise in simulation: no decoherence, no gate errors, and no readout errors affect the AerSimulator.

4. Discussion and Conclusions

The results confirm that Grover's algorithm functions correctly under ideal conditions. The consistently high success probabilities demonstrate that the PhaseOracle correctly marks the target state and that the diffusion operator successfully amplifies its amplitude across all tested qubit counts. This validates both the theoretical predictions of Nielsen & Chuang (2010) and the correctness of the Qiskit implementation.

The most significant finding, however, is the demonstration of the exponential cost of classical quantum simulation. To understand why this is unavoidable, it is necessary to explain what a classical simulator is actually doing. A quantum system of n qubits occupies a superposition of 2^n basis states simultaneously. To represent this exactly, the AerSimulator maintains a state vector of 2^n complex numbers. At 20 qubits, this vector has over one million entries; at 22 qubits, over four million. Every gate operation requires matrix multiplication over this entire state vector. Consequently, both memory usage and computation time scale as $O(2^n)$. This is the fundamental reason why the simulation appeared deceptively fast at small qubit counts (under 1 second up to 12 qubits) but became extremely slow beyond 18 qubits. The transition is not gradual, it is a cliff, because exponential growth is slow to manifest but catastrophic once it does. At 30 qubits, the state vector would require roughly 16 gigabytes of RAM; at 50 qubits, more memory than exists on Earth.

This exponential wall is precisely what motivates the construction of real quantum hardware. A quantum processor of n qubits does not simulate its own quantum state, it physically instantiates it. The hardware is the state vector. Scaling from 22 to 23 qubits requires adding one more physical qubit, not doubling the memory of a classical computer. This is why quantum hardware, despite being far noisier and less reliable than simulation, is the necessary long-term solution for quantum-scale computation.

Comparing the simulation results to Thorvaldson et al. (2025) highlights the complementary roles of simulation and hardware. The silicon processor achieved 95% success at 4 qubits under realistic noise conditions, while simulation achieved 96% at 4 qubits in a noise-free environment. The small difference is meaningful: it represents the cumulative cost of decoherence (quantum states losing coherence before the circuit completes), gate infidelity (imperfect unitary operations), and measurement error on physical hardware. These are not engineering oversights but fundamental physical limitations that grow more severe as circuit depth and qubit count increase. The hardware paper demonstrates that silicon spin qubits are a promising platform because they benefit from mature semiconductor fabrication and exhibit relatively long coherence times, but significant engineering challenges remain before they can scale to the qubit counts where Grover's algorithm would offer a practical advantage.

Classical simulation played an essential role in this project. It allowed the algorithm to be verified, explored, and understood at a level of detail not possible by examining hardware results alone. By inspecting transpiled circuit diagrams for small qubit counts, it was possible to trace exactly how the oracle and diffusion operator are decomposed into elementary gates. The scaling graphs made the exponential cost of simulation viscerally apparent in a way that textbook analysis alone cannot convey. Simulation is therefore valuable not as a substitute for quantum hardware, but as a learning and prototyping tool, precisely the role described in the Qiskit documentation.

Future work could proceed in several directions. First, incorporating a noise model into the AerSimulator (using Qiskit's noise module to mimic realistic gate error rates and decoherence) would allow a more direct quantitative comparison with hardware results. Second, running the algorithm on actual quantum hardware via the IBM Quantum platform would test whether the near-perfect simulation success probabilities degrade in the same pattern observed by Thorvaldson et al. Third, multi-solution variants of Grover's algorithm, where more than one target state exists, could be explored, as the iteration formula changes in that setting. Fourth, alternative amplification strategies such as Quantum Amplitude Estimation could be compared against standard Grover search to investigate whether circuit depth or gate count can be reduced.

5. References

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